MuGyum **Kim**

AI/ML Undergraduate Researcher · Computer Vision & Generative Models

Daejeon, Republic of Korea

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Education

Chungnam National University — *Daejeon, Republic of Korea*

B.Eng. in Artificial Intelligence — *Mar. 2022 – Feb. 2028*

- Major GPA 4.39/4.5 | Cumulative GPA 4.22/4.5

University of Illinois Urbana-Champaign, Grainger College of Engineering — *Urbana-Champaign, IL, USA*

Undergraduate Summer School · AI and Generative Models (Certificate of Recognition) — *Jul. 2026 – Aug. 2026*

- Reimplemented DDPM on MNIST from scratch. Class conditioning raised label agreement from 0.700 to 0.993, and RePaint cut outside-mask error from 0.701 to 0.005.

Publications

MuGyum Kim*, GiWon Kim, ChanJong Park, SungHo Lee†. "Aggregation Sensitivity in Diffusion Inpainting Along a Mean–Median Axis." *IEEE/IEIE International Conference on Consumer Electronics–Asia (ICCE-Asia) 2026*, pp. 396–401. **Oral.**

*Presenting author. †Corresponding author.

Research Experience

DIPLAB, Chungnam National University — *Daejeon, Republic of Korea*

Undergraduate Researcher · Funded by Saemmulter, Inc. — *Sep. 2026 – Oct. 2026*

- Compared TCN and DLSTM groundwater-level forecasts with a persistence baseline on 58 wells, opening the test period only once.
- Found no significant TCN–DLSTM difference after Holm correction and recommended TCN by a pre-specified size rule.
- Delivered the report, an HTML dashboard, and a bit-reproducible package (202 reruns identical).

DIPLAB, Chungnam National University — *Daejeon, Republic of Korea*

Undergraduate Researcher · Funded by ETRI — *Mar. 2026 – Sep. 2026*

- Built a firewall flow-log detector on packet-count-triggered epochs (Scan Detection Sketch) with 28 behavioral features.
- Trained a multi-label MLP for four traffic classes, reaching 94.8% test macro-F1 on source-IP-disjoint splits.
- Ran it on 16.2M unlabeled production epochs and deployed scanning detection as a ClickHouse service.

DIPLAB, Chungnam National University — *Daejeon, Republic of Korea*

Undergraduate Researcher · Satellite Cropland Monitoring, with Contec Co., Ltd. — *Jun. 2023 – Dec. 2023*

- Curated and labeled an 81,000+ sample satellite imagery dataset from Contec for cropland and greenhouse segmentation.
- Trained and compared UNet, SegFormer, HRNet, Hourglass, and DeepLab, reaching 88.9% mIoU against a 60% target.
- Built mask-to-polygon conversion so model outputs match the polygon-format test labels.

DIPLAB, Chungnam National University — *Daejeon, Republic of Korea*

Undergraduate Researcher · CCTV Drunk-Driving Risk Prediction — *May 2023 – Feb. 2024*

- Built and labeled a vehicle pose estimation dataset from CCTV footage for a drunk-driving analysis pipeline.

Skills

Languages & ML Python, C/C++, PyTorch, segmentation-models-pytorch, timm

Libraries & Tools NumPy, Pandas, scikit-learn, OpenCV, ClickHouse, FastAPI, Weights & Biases, Git

Honors & Activities

2022–23 **Academic Excellence Scholarship**, Chungnam National University (2022/2, 2023/1, 2023/2)

2023 **SW-Centered University Activity Scholarship**, Chungnam National University

2026 **Undergraduate Tutor, "AI and Future Society"**, Chungnam National University

2023 **Treasurer, DIVE (AI Club)**, organized a corn seed classification competition